HR Analytics Dashboard

Workforce, Attrition & Retention Analysis

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Power BI

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Project Overview

Background

This project aims to analyze workforce composition, employee attrition, retention pattern, and employee experience using Power BI. The goals is to identify key trends and provide actionable insights to support HR decision-making.

Data Source



HR_Employee_Attrition.csv

(1,471 rows x 37 columns)

Objectives



Understand workforce demographic & compositions



Analyze attrition patterns & risk factors.



Explore retention across tenure, compensation & experience



Provide actionable recommendations for HR

Key Findings - Summary

Workforce

1,470

Attrition Rate

16%

Avg. Tenure

7 years

Retention

84%

Avg. Salary (All)

$6.50K

Workforce

- R&D dominates the workforce (67.2%), followed by Sales (28.7%).
- Employees aged 25–44 form the largest workforce groups.
- The 6–10 years tenure group has the highest employee count (516).

Attrition

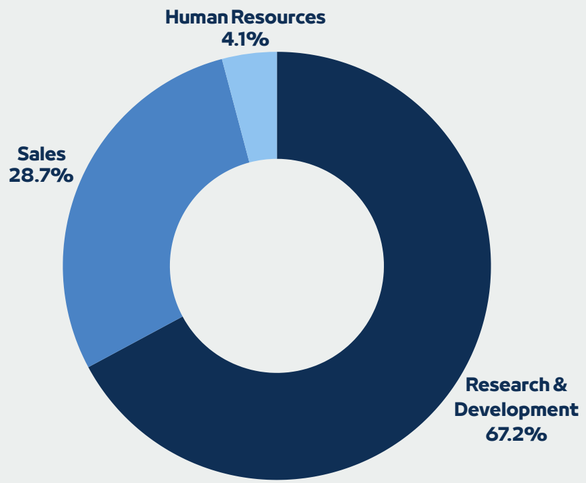
- Attrition is highest among employees aged 18–24 and those with 0–2 years of tenure.
- Sales (21%) and Laboratory Technicians (24%) lead attrition rates.
- Low income, job dissatisfaction, early-tenure, overtime, and longer commutes are key attrition indicators.

Retention & Experience

- Most employees are low risk (66.2%), with only 3.5% at high risk.
- Retention is strongest among longer-tenured employees and those with high job involvement.
- Employee satisfaction and work-life balance show varying retention patterns.

Active Workforce Overview - Detailed

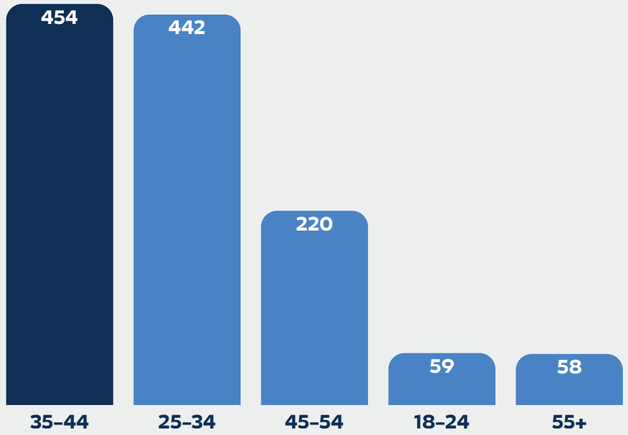
Employee by Department



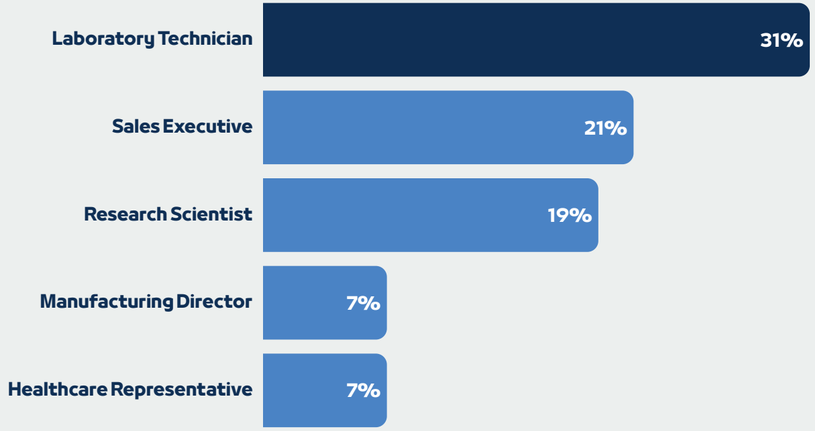
Simulated Hiring Trend



Age Group



Job Role



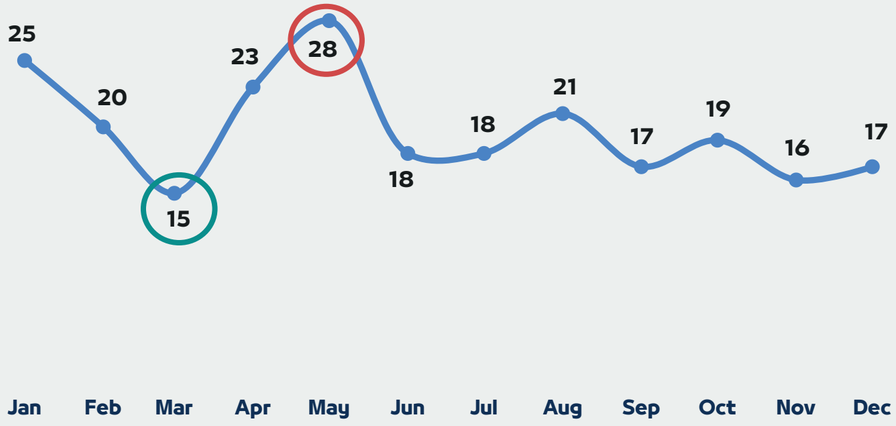
Key Takeaway

R&D dominates the active workforce (67.2%), with employees aged 25–44 forming the largest groups.

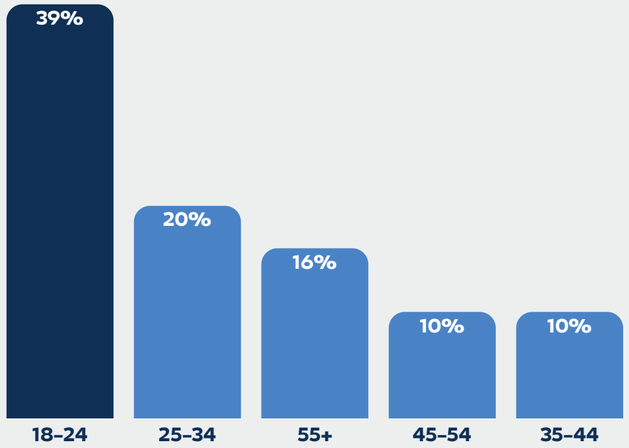
Simulated hiring peaks in May and dips in October.

Attrition Analytics - Detailed

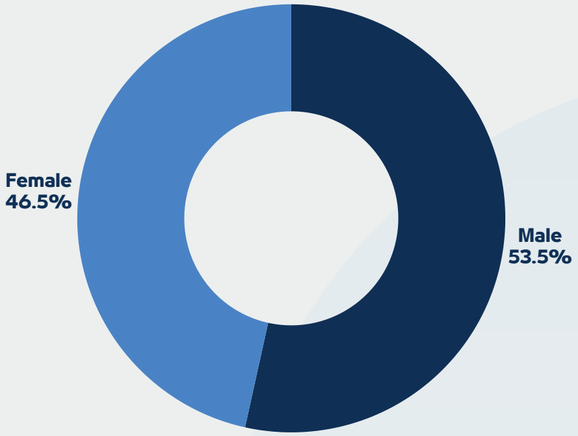
Simulated Attrition Trend



Attrition Rate by Age Group



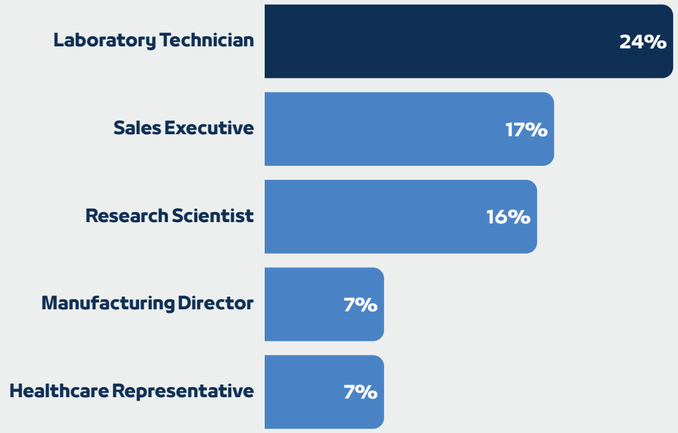
Attrition Rate by Gender



Attrition Rate by Department



Attrition Rate by Job Role



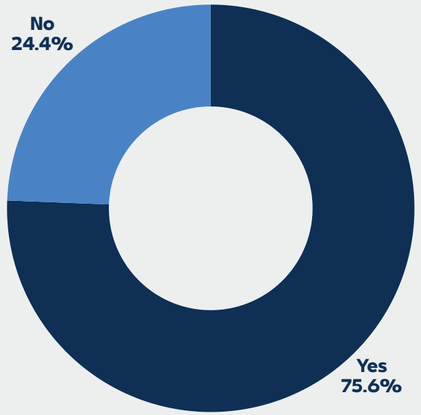
Key Takeaway

Attrition is highest among younger employees (39%) and Laboratory Technicians (24%), with Sales leading by department (21%).

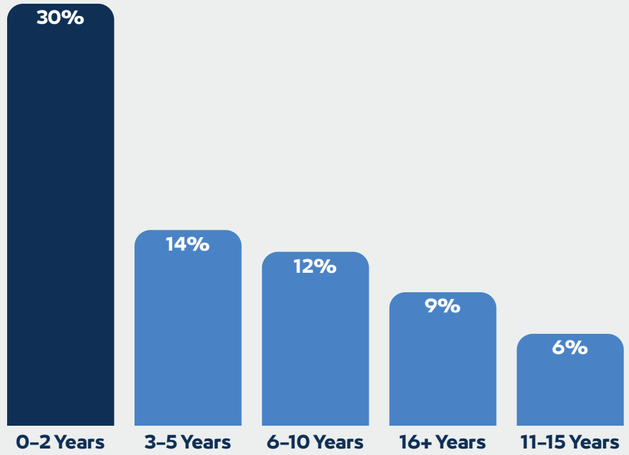
Simulated attrition peaks in May (28) and reaches its lowest in March (15).

Attrition Analytics - Detailed

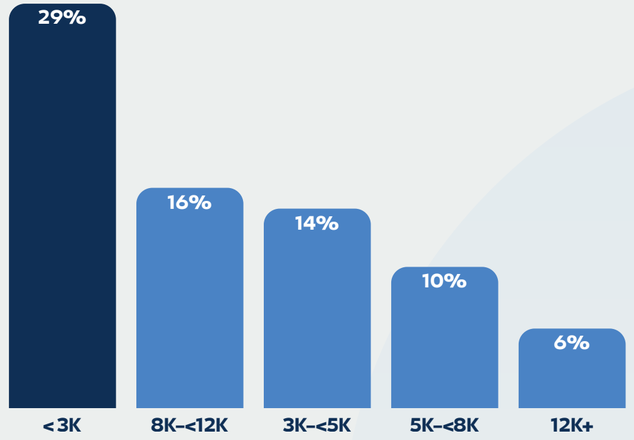
Attrition Rate by Overtime



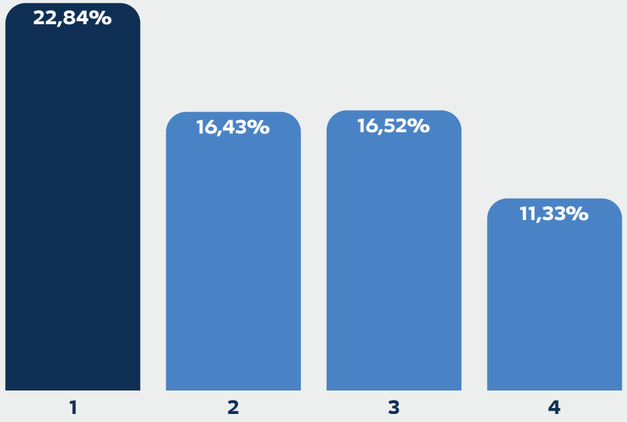
Attrition Rate by Tenure



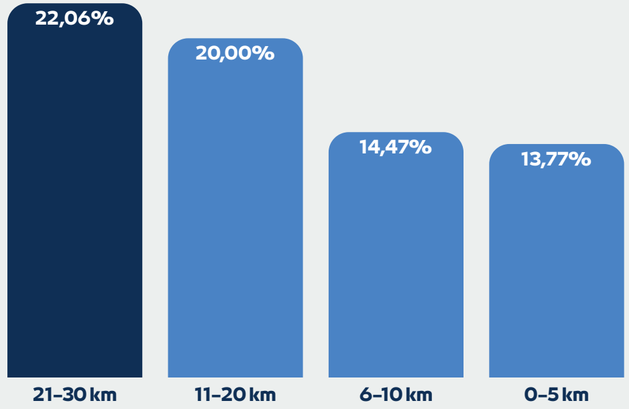
Attrition Rate by Monthly Income



Attrition Rate by Job Satisfaction



Attrition Rate by Distance Home



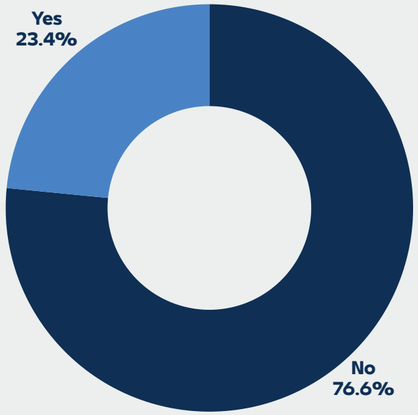
Key Takeaway

Early-tenure employees (0–2 years) and employees earning below 3K show the highest attrition rates.

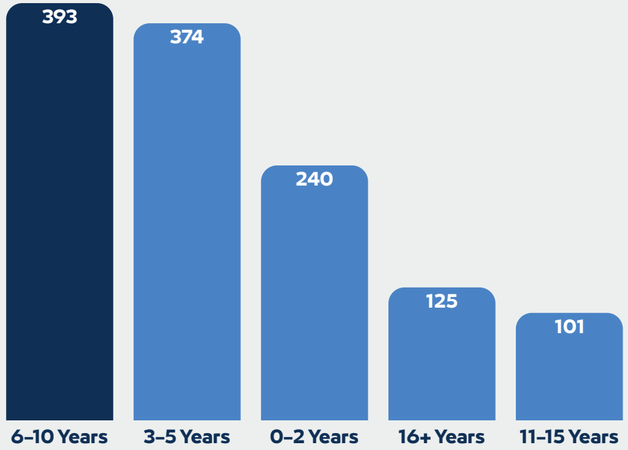
Overtime, low job satisfaction, and longer commuting distances are also relevant attrition indicators. These findings align with previous studies (Marín Díaz et al., 2023; Wang & Liu, 2026).

Retention & Employee Experience - Detailed

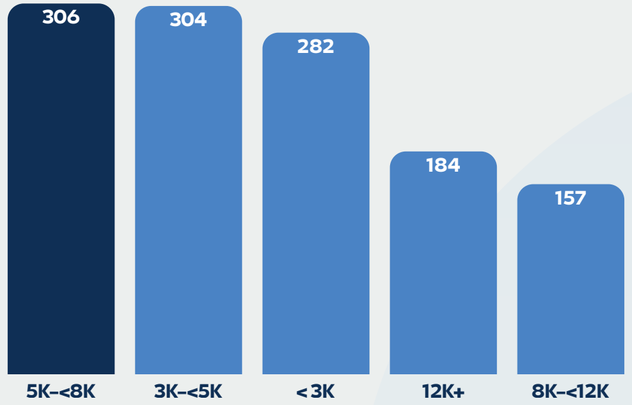
Retention by Overtime



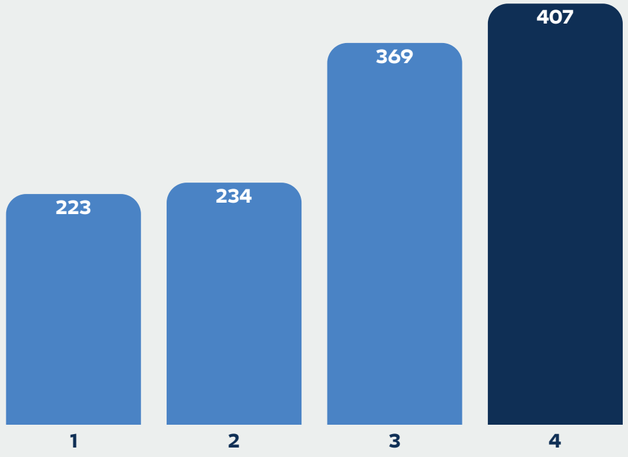
Retained Employees by Tenure



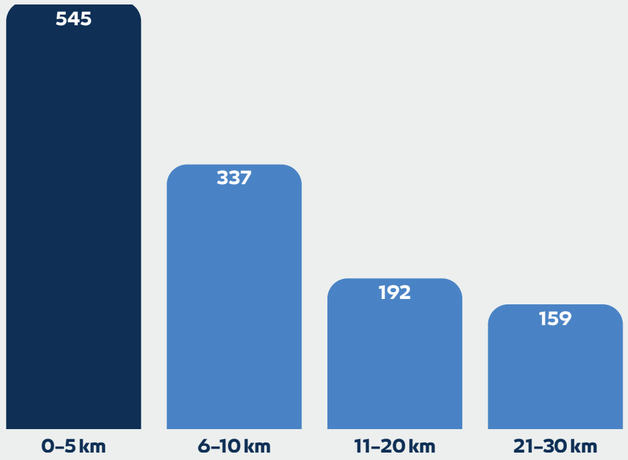
Retained Employees by Income



Retained Employees by Job Satisfaction



Retained Employees by Distance Home



Key Takeaway

Most retained employees do not work overtime (76.6%), have 6–10 years of tenure (393 employees), and earn 5K–<8K (306 employees).

Retained employee counts are highest at job satisfaction level 4 (407) and among those living within 0–5 km of work (545).

Retention & Employee Experience - Detailed

Key Experience Metrics

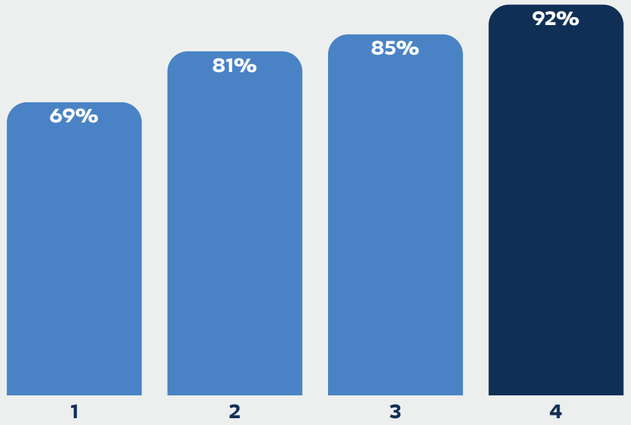
Avg. of Job Satisfaction

2.73

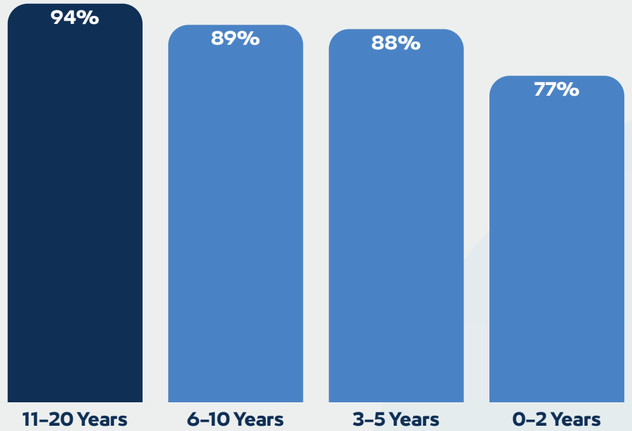
Avg. Work Life Balance

2.76

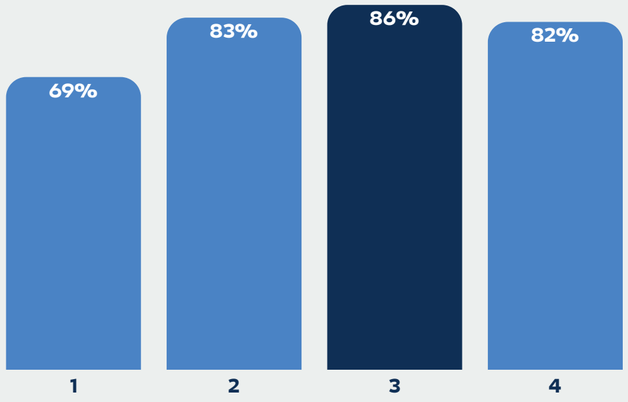
Retention by Job Involvement



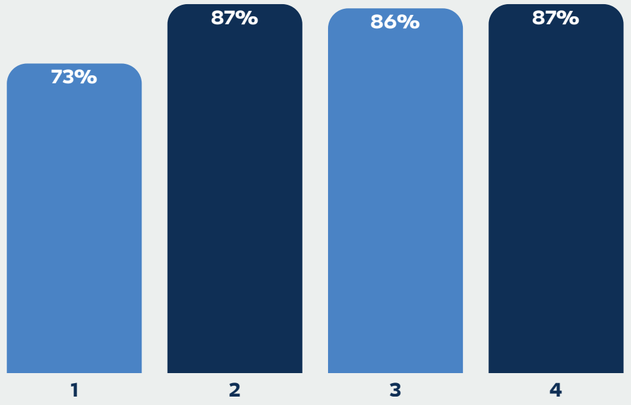
Retention by Years Current Role



Retention by Work Life Balance



Retention by Environment

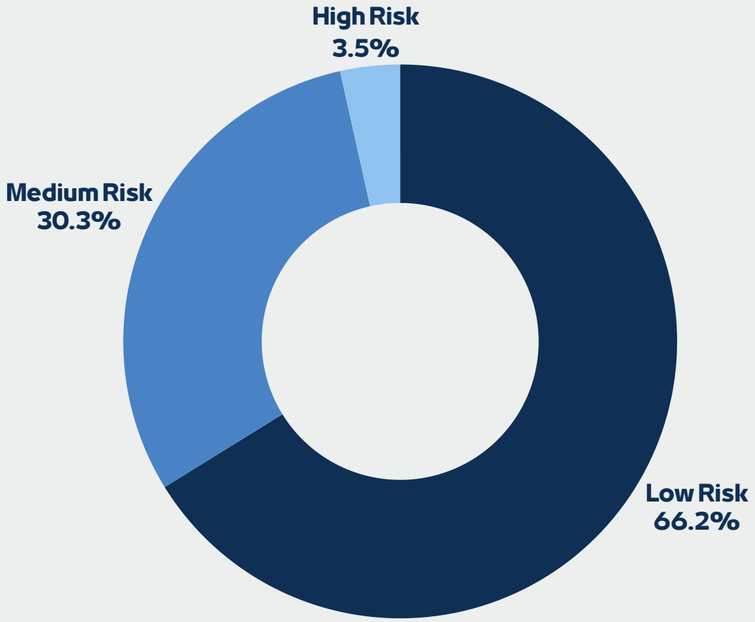


Key Takeaway

Retention is highest among employees with longer tenure in their current role (94%) and high job involvement (92%).

Work-life balance and environment satisfaction show varying retention patterns, with relatively small differences across most levels.

Predicted Attrition



Top 5 Key Drivers of Attrition

Rank	Feature	Importance
1	Overtime	18%
2	Tenure	15%
3	Monthly Income	12%
4	Job Satisfaction	11%
5	Distance from Home	8%

Key Takeaway

Most employees fall into the low-risk category (66.2%), while 3.5% are classified as high risk. Overtime, tenure, monthly income, job satisfaction, and commuting distance emerge as key attrition predictors.

Key Insights

01



Workforce Concentration

R&D represents the largest workforce segment (67.2%), followed by Sales (28.7%). Employees aged 25–44 form the largest age groups, while younger employees aged 18–24 show the highest attrition rate (39%).

02



Hiring & Attrition Trends

Simulated hiring peaks in May (119) and reaches its lowest in January (80). Simulated attrition also peaks in May (28), while March records the lowest (15), highlighting similar monthly peaks in both trends.

03



Department & Job Role Attrition

Sales has the highest departmental attrition rate (21%), while Laboratory Technicians record the highest job-role attrition rate (24%). Attrition is concentrated among Sales Executives and Sales Representatives in Sales, and Laboratory Technicians and Research Scientists in R&D.

04



Key Attrition Factors

Early-tenure employees (30%) and those earning below 3K (29%) show the highest attrition. Low job satisfaction (22.84%), longer commutes (22.06%), and overtime (75.6% of attrition cases) are also key attrition indicators.

05



Retention & Employee Experience

The workforce shows relatively low attrition risk (66.2% low risk; 3.5% high risk). Retention is strongest among longer-tenured employees and those with high job involvement. Most retained employees do not work overtime, while job satisfaction, work-life balance, and environment satisfaction show generally positive but varying patterns.

Recommendations



Stengthen Early-Career Retention

ST
- Improve onboarding and mentorship.
- Provide regular feedback for employees aged 18–24 and those with 0–2 years of tenure.

LT
- Establish clear career development paths.
- Create structured learning and development programs.



Optimize Workforce Planning

ST
- Review staffing needs across departments and job roles.
- Monitor monthly hiring and attrition trends.

LT
- Improve workforce allocation based on business needs.
- Develop long-term workforce planning strategies.



Address Department & Role-Specific Attrition

ST
- Investigate attrition among Sales Executives, Sales Representatives, and Laboratory Technicians.
- Review workload, overtime, and compensation among high-attrition roles.

LT
- Develop targeted retention strategies for high-attrition departments and roles.
- Evaluate compensation and workload policies.



Enhance Employee Experience

ST
- Monitor job satisfaction and employee engagement.
- Identify concerns related to work-life balance and working conditions.

LT
- Strengthen employee engagement initiatives.
- Support career growth, internal mobility, and a positive work environment.

SH = Short-Term **LT** = Long-Term

Building a stronger workforce through targeted people strategies.

References

- Marín Díaz et al. (2023). Analyzing Employee Attrition Using Explainable AI for Strategic HR Decision-Making. Mathematics.
- Wang & Liu (2026). Job Satisfaction, Compensation, and Attrition Risk: Machine Learning Evidence from the IBM HR Analytics Benchmark Dataset. Frontiers in Psychology

HR Analytics Dashboard

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